

Section 5.4

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125010006 T03

$$1. \int_0^2 x(x-3) dx = \int_0^2 (x^2 - 3x) dx = \left[\frac{x^3}{3} - \frac{3x^2}{2} \right]_0^2 = \left(\frac{8}{3} - 6 \right) - (0 - 0) = -\frac{10}{3}$$

$$3. \int_{-2}^2 \frac{3}{(x+3)^4} dx = \left[\frac{3}{(x+3)^3} \cdot \frac{1}{-3} \right]_{-2}^2 = \left[-\frac{1}{(x+3)^3} \right]_{-2}^2 = \left(-\frac{1}{125} + 1 \right) = \frac{124}{125}$$

$$4. \int_1^2 x^{299} dx = \left[\frac{x^{300}}{300} \right]_1^2 = \frac{1}{300} - \left(\frac{1}{300} \right) = 0$$

$$8. \int_1^{1/3} x^{-6/5} dx = \left[\frac{x^{-1/5}}{-1/5} \right]_1^{1/3} = \left[\frac{-5}{x^{1/5}} \right]_1^{1/3} = \left(-\frac{5}{2} + 5 \right) = \frac{5}{2}$$

$$9. \int_0^{\pi/3} 2 \sec^2 x dx = \left[2 \tan x \right]_0^{\pi/3} = \left(2 \cdot \frac{1}{\sqrt{3}} - 0 \right) = \frac{2}{\sqrt{3}}$$

$$10. \int_0^{\pi} (1 + \cos x) dx = \left[x + \sin x \right]_0^{\pi} = (\pi + 0) - (0 + 0) = \pi$$

$$12. \text{Let } v = \cos u \rightarrow dv = -\sin u du$$

$$\therefore \int_{\pi/3}^{\pi/2} 4 \frac{\sin u}{\cos^2 u} du = \int_{\pi/3}^{\pi/2} -\frac{4}{v^2} dv = \int_{\pi/3}^{\pi/2} \frac{4}{v^2} dv = \left(-\frac{4}{v} \right)_{\pi/3}^{\pi/2} = \frac{8}{\pi} - 4 = \frac{8\pi - 12}{3}$$

$$17. \int_0^{\pi/2} \sin 2x dx = \left[-\frac{\cos 2x}{2} \right]_0^{\pi/2} = -\frac{1}{2\sqrt{2}} + \frac{1}{2} = \frac{2 - \sqrt{2}}{4}$$

$$22. \int_{-3}^2 \frac{y^5 - 2y}{y^4} dy = \int_{-3}^2 \left(y - \frac{2}{y} \right) dy = \left[\frac{y^2}{2} + \frac{2}{y} \right]_{-3}^2 = \left(-\frac{1}{2} - 2 \right) - \left(-9 - \frac{2}{3} \right) = \frac{22}{3}$$

$$24. \int_1^8 \frac{(x^{1/3} + 1)(2 - x^{2/3})}{x^{1/3}} dx = \int_1^8 \frac{-x - x^{2/3} + 2x^{1/3} + 2}{x^{1/3}} dx = \int_1^8 (-x^{2/3} - x^{1/3} + 2 + 2x^{-1/3}) dx$$

$$= \left[-\frac{3}{5} x^{5/3} - \frac{3}{4} x^{4/3} + 2x + 3x^{2/3} \right]_1^8$$

$$= \left(-\frac{96}{5} - 12 + 16 + 12 \right) - \left(-\frac{3}{5} - \frac{3}{4} + 2 + 3 \right) = -\frac{137}{20}$$

$$27. \int_{-4}^4 |x| dx = \int_{-4}^0 -x dx + \int_0^4 x dx$$

$$= \left[-\frac{x^2}{2} \right]_{-4}^0 + \left[\frac{x^2}{2} \right]_0^4 = 8 + 8 = 16$$

$$28. \int_0^{\pi} \frac{1}{2} (\cos x + \cos x) dx = \int_0^{\pi} \frac{1}{2} (\cos x + \cos x) dx + \int_{\pi/2}^{\pi} \frac{1}{2} (\cos x - \cos x) dx$$

$$= \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} 0 dx$$

$$= \left[\sin x \right]_0^{\pi/2} + 0 = 1 - 0 + 0 = 1$$

$$30. \int_1^{\pi^2} \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \left[-2 \cos \sqrt{x} \right]_1^{\pi^2} = -2 \cos \pi + 2 \cos 1 = 2 + 2 \cos 1$$

$$40. \frac{dy}{dx} = \frac{d}{dx} \left(\int_1^x \frac{1}{t} dt \right) = \frac{1}{x}$$

$$44. \frac{dy}{dx} = \frac{d}{dx} \left(\int_0^x (t^2 + 1)^{10} dt \right) = 3 \left(\int_0^x (t^2 + 1)^{10} dt \right)^2 \cdot \frac{1}{dx} \int_0^x (t^2 + 1)^{10} dt$$

$$= 3 \left(\int_0^x (t^2 + 1)^{10} dt \right)^2 \cdot (x^2 + 1)^{10}$$

$$46. \frac{dy}{dx} = \frac{d}{dx} \int_{\tan x}^0 \frac{dt}{1+t^2} = \frac{d}{dx} \left(-\int_0^{\tan x} \frac{1}{1+t^2} dt \right) = \frac{1}{1+\tan^2(x)} \left(\frac{d}{dx} \tan x \right) = \frac{1}{\sec^2 x} \cdot \sec^2(x) = 1$$

$$61. \int_{-b/2}^{b/2} h - \left(\frac{4h}{b^2} \right) x^2 dx = \left[hx - \frac{4h}{3b^2} x^3 \right]_{-b/2}^{b/2} = \left(\frac{hb}{2} - \frac{hb}{6} \right) - \left(-\frac{hb}{2} + \frac{hb}{6} \right)$$

$$= \frac{2hb}{3}$$

$$63. c(100) - c(1) = \int_1^{100} \frac{1}{2\sqrt{x}} dx = \left[\sqrt{x} \right]_1^{100} = 9$$

$$68. \int_0^{\pi} f(x) dx = x \cos \pi x \rightarrow f(x) = \frac{d}{dx} \left(\int_0^x f(t) dt \right) = \cos(\pi x) - \pi x \sin(\pi x) \therefore f(4) = 1 - \pi(4)(0) = 1$$

$$71. a) \text{ True } \left(\frac{dy}{dx} = f(x), f \text{ differentiable} \rightarrow g \text{ differentiable} \rightarrow g \text{ continuous} \right)$$

b) True

c) True $(g'(1) = f(1) = 0, g''(1) = f'(1) > 0 \rightarrow g \text{ has horizontal tangent at } x=1)$

d) False

e) True $(g'(1) = 0, g''(1) > 0)$

f) False $(g''(x) > 0 \text{ for all } x)$

g) True $(g''(x) > 0 \forall x, g'(1) = 0)$

Section 5.4 (Continue)

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73. a) Velocity at $t=5$ equals $f(5)=2$ ($f \rightarrow$ velocity function)
 b) Negative (Slope of $f =$ acceleration)
 c) $S(3) = \int f(x) dx =$ area under graph of $f(x)$ from $t=0$ to $t=3 = \frac{1}{2} \cdot 3 \cdot 3 = \frac{9}{2}$
 d) $t=6$ s (After this, $v < 0$ means moving backwards)
 e) $a(t) = 0$ when the slope of $f(x)$ is zero, or at $t=4$ s, 7 s
 f) Towards origin: From $t=6$ s to $t=9$ s, since $f(t) < 0$
 Away from origin: From $t=0$ s to $t=6$ s, since $f(t) > 0$
 g) Positive axis (Total area under curve from $t=0$ to $t=9$ is positive)

Section 5.5

18. Let $u = 5x+4 \rightarrow du = 5 dx \therefore \int \frac{1}{\sqrt{5x+4}} dx = \int \frac{1}{5} \cdot \frac{1}{\sqrt{u}} du = \frac{1}{5} \cdot 2\sqrt{u} + C = \frac{2}{5} \sqrt{5x+4} + C$
 20. Let $u = 7-3y^2 \rightarrow du = -6y dy \therefore \int 3y \sqrt{7-3y^2} dy = \int -\frac{1}{2} \sqrt{u} du = -\frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C = -\frac{1}{3} (7-3y^2)^{3/2} + C$
 21. Let $u = 1+\sqrt{x} \rightarrow du = \frac{1}{2\sqrt{x}} dx \therefore \int \frac{1}{\sqrt{x}(1+\sqrt{x})} dx = \int 2 \cdot \frac{1}{u} du = -\frac{2}{u} + C = -\frac{2}{1+\sqrt{x}} + C$
 22. Let $u = \sqrt{\sin x} \rightarrow du = \frac{1}{2\sqrt{\sin x}} \cdot \cos x dx \therefore 2u^2 du = \sqrt{\sin x} \cdot \cos x dx, \cos^2 x = 1 - \sin^2 x = 1 - (\sin x)^4 = 1 - u^4$
 $\therefore \int \sqrt{\sin x} \cdot \cos x dx = \int 2u^2(1-u^4) du = \int (2u^2 - 2u^6) du = \frac{2}{3} u^3 - \frac{2}{7} u^7 + C = \frac{2}{3} (\sin x)^{3/2} - \frac{2}{7} (\sin x)^{7/2} + C$
 26. Let $u = \tan \frac{x}{2} \rightarrow du = \frac{1}{2} \sec^2 \frac{x}{2} dx \therefore \int \tan^3 \frac{x}{2} \sec^2 \frac{x}{2} dx = \int 2u^3 du = \frac{u^4}{2} + C = \frac{1}{2} \tan^4 \frac{x}{2} + C$
 31. Let $u = \cos(2t+1) \rightarrow du = -2 \sin(2t+1) \therefore \int \frac{\sin(2t+1)}{\cos^2(2t+1)} dx = \int -\frac{1}{2u^2} du = \frac{1}{2u} + C = \frac{1}{2 \cos(2t+1)} + C$
 38. Let $u = 1 - \frac{1}{x} \rightarrow du = \frac{1}{x^2} dx \therefore \int \sqrt{\frac{x-1}{x^2}} dx = \int \frac{1}{x} \cdot \sqrt{1-\frac{1}{x}} dx = \int -\sqrt{u} du = -\frac{2}{3} u^{3/2} + C = -\frac{2}{3} (1-\frac{1}{x})^{3/2} + C$
 42. Let $u = x^2 - 1 \rightarrow du = 2x dx \therefore \int \frac{x^3}{x^2-1} dx = \int \frac{x^2}{x^2-1} dx = \int \frac{1}{2\sqrt{u}} du = \frac{1}{\sqrt{u}} + C = \frac{1}{\sqrt{x^2-1}} + C$
 43. Let $u = x-1 \rightarrow du = dx \therefore \int x(x-1)^{10} dx = \int (u+1)u^{10} du = \int (u^{11} + u^{10}) du = \frac{u^{12}}{12} + \frac{u^{11}}{11} + C = \frac{(x-1)^{12}}{12} + \frac{(x-1)^{11}}{11} + C$
 46. Let $u = x-5 \rightarrow du = dx \therefore \int (x+5)(x-5)^{1/3} dx = \int (u+10)u^{1/3} du = \int (u^{4/3} + 10u^{1/3}) du = \frac{3}{7} u^{7/3} + \frac{15}{2} u^{4/3} + C$
 $= \frac{3}{7} (x-5)^{7/3} + \frac{15}{2} (x-5)^{4/3} + C$
 56. Let $u = x^2+8 \rightarrow du = 2x dx \therefore y = \int 4x(x^2+8)^{-1/3} dx = \int 2u^{-1/3} du = 3u^{2/3} + C = 3(x^2+8)^{2/3} + C$
 $\therefore 0 = y(0) = 3(0^2+8)^{2/3} + C \rightarrow C = -12 \therefore y = 3(x^2+8)^{2/3} - 12$
 62. $a = \frac{d^2s}{dt^2} = \pi^2 \cos \pi t \rightarrow v = \frac{ds}{dt} = \pi \sin \pi t + v_0$
 $\therefore 8 = v(0) = \pi \sin 0 + v_0 \rightarrow v_0 = 8 \therefore \frac{ds}{dt} = \pi \sin \pi t + 8$
 $\therefore s = -\cos \pi t + 8t + s_0 \therefore 0 = s(0) = -\cos 0 + 8(0) + s_0 \rightarrow s_0 = 1$
 $\therefore s(t) = -\cos \pi t + 8t + 1$
 $s(1) = -\cos \pi + 8 + 1 = 10 \text{ m}$

Section 5.6

8. a) Let $u = (1+v^{3/2}) \rightarrow du = \frac{3}{2} v dv \therefore \int_0^1 \frac{10\sqrt{v}}{(1+v^{3/2})^2} dv = \int_1^4 \frac{20}{3u^2} du = \left[-\frac{20}{3u} \right]_1^4 = -\frac{20}{12} + \frac{20}{3} = 5$
 b) $\int_1^4 \frac{10\sqrt{v}}{(1+v^{3/2})^2} dv = \int_4^1 \frac{20}{3u^2} du = \left[-\frac{20}{3u} \right]_4^1 = \frac{20}{12} - \frac{20}{243} = \frac{385}{243}$
 11. a) Let $u = 5t+4 \rightarrow t = \frac{1}{5}(u-4), du = 5dt \rightarrow \int_0^9 t \sqrt{4+5t} dt = \int_4^49 \frac{1}{25} (u-4) \sqrt{u} du = \frac{1}{25} \int_4^49 (u^{3/2} - 4u^{1/2}) du = \frac{1}{25} \left[\frac{2}{5} u^{5/2} - \frac{8}{3} u^{3/2} \right]_4^49$
 $= \frac{1}{25} \left[\frac{2}{5} (21)^{5/2} - \frac{8}{3} (19)^{3/2} \right] = \frac{506}{375}$

11. b) $\int_1^4 \frac{1}{t} \sqrt{4+5t} dt = \frac{1}{25} \left[\frac{2}{3} u^{5/2} - \frac{8}{3} u^{3/2} \right]_9^{49} = \frac{1}{25} \left[\left(\frac{3364}{5} - \frac{486}{5} \right) - \left(\frac{2744}{3} - \frac{216}{3} \right) \right] = \frac{86744}{375}$

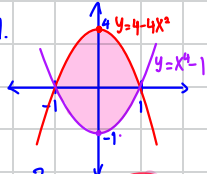
12. a) Let $u = 1 - \cos 3t \rightarrow du = 3 \sin 3t \therefore \int_0^{\pi/6} (1 - \cos 3t) \sin 3t dt = \int_0^1 \frac{1}{3} u du = \left[\frac{1}{6} u^2 \right]_0^1 = \frac{1}{6}$

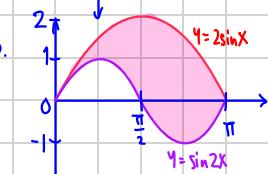
b) $\int_{\pi/6}^{\pi/3} (1 - \cos 3t) \sin 3t dt = \int_1^2 \frac{1}{3} u du = \left[\frac{1}{6} u^2 \right]_1^2 = \frac{1}{2}$

21. Let $u = 4y^3 - y^2 + 4y + 1 \rightarrow du = (12y^2 - 2y + 4) dy \therefore \int_0^1 (4y^3 - y^2 + 4y + 1) (12y^2 - 2y + 4) dy = \int_1^8 u^{-2/3} du = \left[3 \sqrt[3]{u} \right]_1^8 = 3$

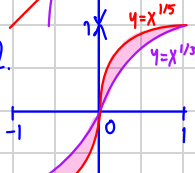
23. Let $u = \theta^{3/2} \rightarrow du = \frac{3}{2} \sqrt{\theta} d\theta \therefore \int_0^{\pi/4} \sqrt{\theta} \cos^2(\theta^{3/2}) d\theta = \int_0^{\pi/2} \frac{2}{3} \cos^2 u du = \frac{2}{3} \int_0^{\pi/2} \frac{1 + \cos 2u}{2} du = \frac{1}{3} \left[\frac{u}{2} + \frac{\sin 2u}{4} \right]_0^{\pi/2} = \frac{1}{3} \left(\frac{\pi}{2} \right) = \frac{\pi}{6}$

25. Since $y(-x) = -x\sqrt{4-(-x)^2} = -x\sqrt{4-x^2} = -y(x) \rightarrow y$ is odd function $\therefore \int_{-2}^2 y(x) dx = 0$

59.  $Area = \int_{-1}^1 [(4 - 4x^2) - (x^4 - 1)] dx = \int_{-1}^1 (-x^4 - 4x^2 + 5) dx = \left[-\frac{x^5}{5} - \frac{4}{3}x^3 + 5x \right]_{-1}^1 = -\frac{2}{5} - \frac{8}{3} + 10 = \frac{104}{15}$

63.  $Area = \int_0^{\pi} (2 \sin x - \sin 2x) dx = \left[-2 \cos x + \frac{\cos 2x}{2} \right]_0^{\pi} = -2(-1) + 2(1) + \frac{1}{2} - \frac{1}{2} = 4$

71.  $Area = \int_{-1}^0 (x^3 - x) dx + \int_0^1 (x - x^3) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = 0 - \left(-\frac{1}{4} \right) + \left(\frac{1}{2} - 0 \right) = \frac{1}{2}$

72.  $Area = \int_{-1}^0 (x^{1/3} - x^{1/5}) dx + \int_0^1 (x^{1/5} - x^{1/3}) dx = 2 \int_0^1 (x^{1/5} - x^{1/3}) dx = 2 \left[\frac{5}{6} x^{6/5} - \frac{3}{4} x^{4/3} \right]_0^1 = 2 \left(\frac{5}{6} - \frac{3}{4} \right) = \frac{1}{6}$

78. Shaded area = $\int_0^1 2\sqrt{y} dy + \int_1^2 \underbrace{(3-y) - (y-1)^2}_{-y^2+y+2} dy = \left[\frac{4}{3} y^{3/2} \right]_0^1 + \left[-\frac{y^3}{3} + \frac{y^2}{2} + 2y \right]_1^2 = \frac{4}{3} + \left(-\frac{8}{3} + \frac{3}{2} + 2 \right) = \frac{5}{6}$

82. Sometimes true. We have to know which function (f or g) is greater than one another in each sub-interval of $[a, b]$. Using that formula could occur some x such that $f(x) < g(x)$ which does not make sense when finding area under graph, so we could change from $\int_a^b (f(x) - g(x)) dx$ to $\int_a^b |f(x) - g(x)| dx$.

83. $u = 2x \rightarrow du = 2 dx \Rightarrow \int_2^6 \frac{\sin^4 u}{u} du = \int_2^6 \frac{\sin^4 u}{u} du = F(6) - F(2)$

84. Let $u = 1 - x \rightarrow du = -dx \therefore \int_0^1 f(u) du = \int_1^0 f(1-x) (-dx) = \int_0^1 f(1-x) dx$

85. a) $\int_{-1}^0 f(x) dx = - \int_0^{-1} f(x) dx = -3$

b) $\int_0^1 f(x) dx = \int_0^1 f(x) dx = 3$

86. a) Let $u = -x \rightarrow du = -dx, f(u) = -f(x) \therefore \int_{-2}^2 f(x) dx = \int_2^{-2} (-f(u)) (-du) = \int_2^{-2} f(u) du = - \int_{-2}^2 f(x) dx$
 $\therefore \int_{-2}^2 f(x) dx = 0 \rightarrow \int_{-2}^2 f(x) dx = 0$

b) Use $f(x) = \sin x, a = \frac{\pi}{2} \rightarrow \int_{-\pi/2}^{\pi/2} \sin x dx = [-\cos x]_{-\pi/2}^{\pi/2} = -0 - (-0) = 0$

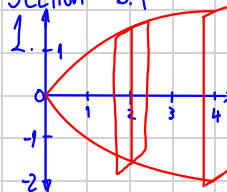
Section 5.6 (Continue)

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87. Let $u = a - x \rightarrow du = -dx$, $I = \int_0^a \frac{f(x) dx}{f(x) + f(a-x)} = \int_a^0 \frac{f(a-u) (-du)}{f(a-u) + f(u)} = \int_0^a \frac{f(a-x) dx}{f(a-x) + f(x)}$

$$\therefore 2I = \int_0^a \frac{f(x) dx}{f(x) + f(a-x)} + \int_0^a \frac{f(a-x) dx}{f(a-x) + f(x)} = \int_0^a 1 dx = a \quad \therefore I = \frac{a}{2}$$

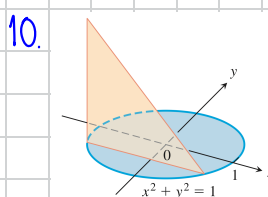
Section 6.1

1. 

$$V = \int_0^4 (2\sqrt{x})^2 dx$$

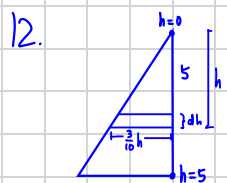
$$= \int_0^4 (2\sqrt{x})^2 dx$$

$$= \int_0^4 4x dx = \left[\frac{4x^2}{2} \right]_0^4 = 32$$

10. 

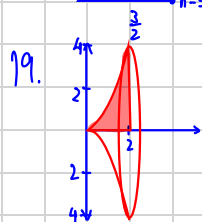
$$A(y) = \frac{1}{2} \cdot (2x)^2 = 2x^2 = 2(1-y^2) = 2-2y^2$$

$$\therefore V = \int_{-1}^1 A(y) dy = \int_{-1}^1 (2-2y^2) dy = \left(2y - \frac{2y^3}{3} \right)_{-1}^1 = \frac{8}{3}$$

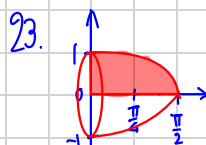
12. 

$$A(h) = \left(2 \left(\frac{3}{10} h \right) \right)^2 = \left(\frac{3}{5} h \right)^2$$

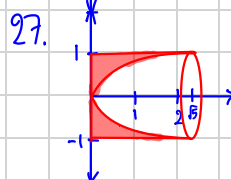
$$V = \int_0^5 A(h) dh = \int_0^5 \left(\frac{3}{5} h \right)^2 dh = \int_0^5 \frac{9}{25} h^2 dh = \left[\frac{3h^3}{25} \right]_0^5 = 15$$

19. 

$$V = \int_{-2}^2 \pi (x^2)^2 dx = \int_{-2}^2 \pi x^4 dx = \left[\frac{\pi x^5}{5} \right]_{-2}^2 = \frac{32\pi}{5}$$

23. 

$$V = \int_0^{\pi/2} \pi (\cos x)^2 dx = \int_0^{\pi/2} \pi \cos x dx = \pi [\sin x]_0^{\pi/2} = \pi$$

27. 

$$X = \sqrt{5} y^2$$

$$V = \int_{-1}^1 \pi (\sqrt{5} y^2)^2 dy = \int_{-1}^1 \pi 5 y^4 dy = \pi \left[\frac{y^5}{5} \right]_{-1}^1 = 2\pi$$

14. 1) Both cones have altitude of 12

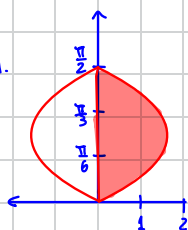
2) Both cones have cross section as circle with diameter $\frac{x}{2}$ (radius $\frac{x}{4}$) at every altitude

3) Cavalieri's Principle: identical cross section, equal altitude

$h=12, r=3$ (second cone)

→ same volume

29.



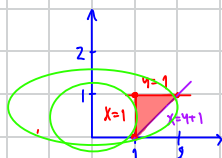
$$V = \int_0^{\pi/2} \pi (\sqrt{2 \sin 2y})^2 dy = \int_0^{\pi/2} 2\pi \sin 2y dy = 2\pi [-\cos 2y]_0^{\pi/2} = 2\pi(2) = 4\pi$$

37.



$$\begin{aligned} V &= \int_{-1}^2 \pi [(x+3)^2 - (x^2+1)^2] dx \\ &= \int_{-1}^2 \pi [x^2+6x+9 - x^4+2x^2-1] dx \\ &= \int_{-1}^2 \pi [-x^4+3x^2+6x+8] dx \\ &= \pi \left[-\frac{x^5}{5} + x^3 + 3x^2 + 8x \right]_{-1}^2 = \frac{148}{5} + \frac{29}{5} = \frac{177}{5} \end{aligned}$$

41.



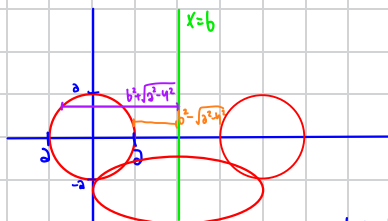
$$\begin{aligned} V &= \int_0^1 \pi [(y+1)^2 - 1^2] dy \\ &= \int_0^1 \pi [y^2+2y] dy = \left[\frac{y^3}{3} + y^2 \right]_0^1 = \frac{4}{3} \end{aligned}$$

43.



$$\begin{aligned} y &= x^2 \rightarrow x = \sqrt{y} \\ \therefore V &= \int_0^4 \pi (2^2 - (\sqrt{y})^2) dy = \int_0^4 \pi (4 - y) dy = \left[4\pi y - \frac{\pi y^2}{2} \right]_0^4 = 16\pi - 8\pi = 8\pi \end{aligned}$$

51.



$$\begin{aligned} V &= \int_{-a}^a \pi ((b + \sqrt{a^2 - y^2})^2 - (b - \sqrt{a^2 - y^2})^2) dy \\ &= \int_{-a}^a \pi (4b\sqrt{a^2 - y^2}) dy \\ &= 4\pi b \int_{-a}^a \sqrt{a^2 - y^2} dy = 4\pi b \left(\frac{\pi a^2}{2} \right) = 2\pi^2 a^2 b \end{aligned}$$

$$59. \text{ From } \int_a^b \pi (f(x)+1)^2 dx = 8\pi \rightarrow \int_a^b \pi [f(x)^2 + 2f(x) + 1] dx = 8\pi$$

$$\int_a^b \pi (f(x)^2 + 2f(x) + 1) dx = 8 \quad (1)$$

$$\int_a^b \pi (f(x))^2 dx = 4\pi \rightarrow \int_a^b \pi (f(x))^2 dx = 4 \quad (2)$$

$$(1) - (2): \int_a^b \pi (2f(x) + 1) dx = 4$$

$$\therefore R = \int_a^b f(x) dx = \frac{3}{2}$$